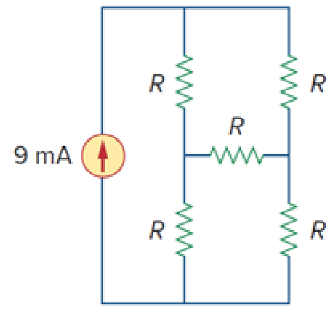
Problem

Design a problem to help other students better understand wye-delta transformations using Fig. 2.114.

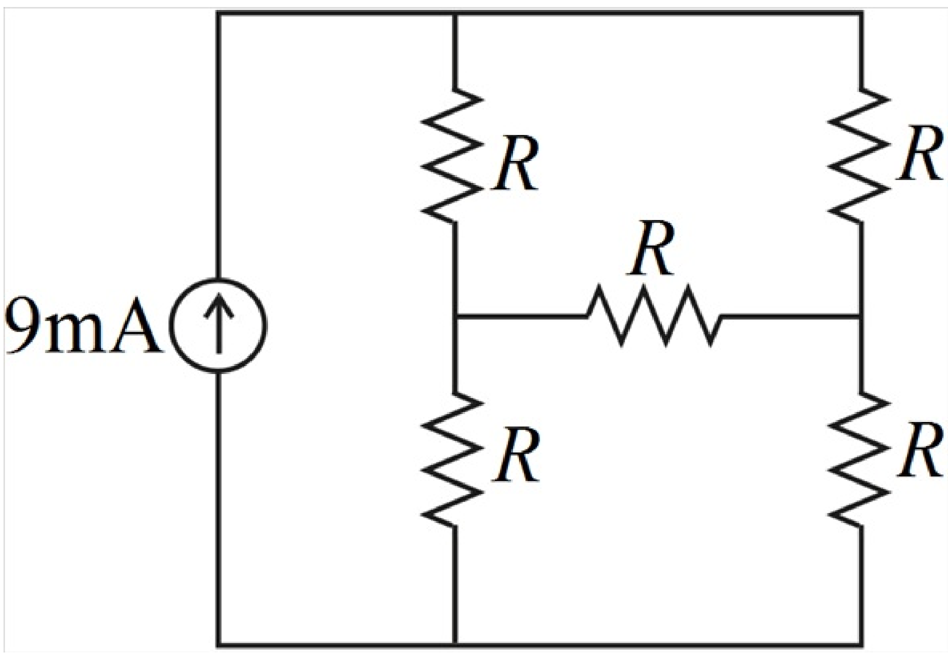
Figure 2.114:



Step-by-step solution

Step 1 of 10

The given circuit is



Step 2 of 10

To find the equivalent resistance as seen from the terminals of current source, we use

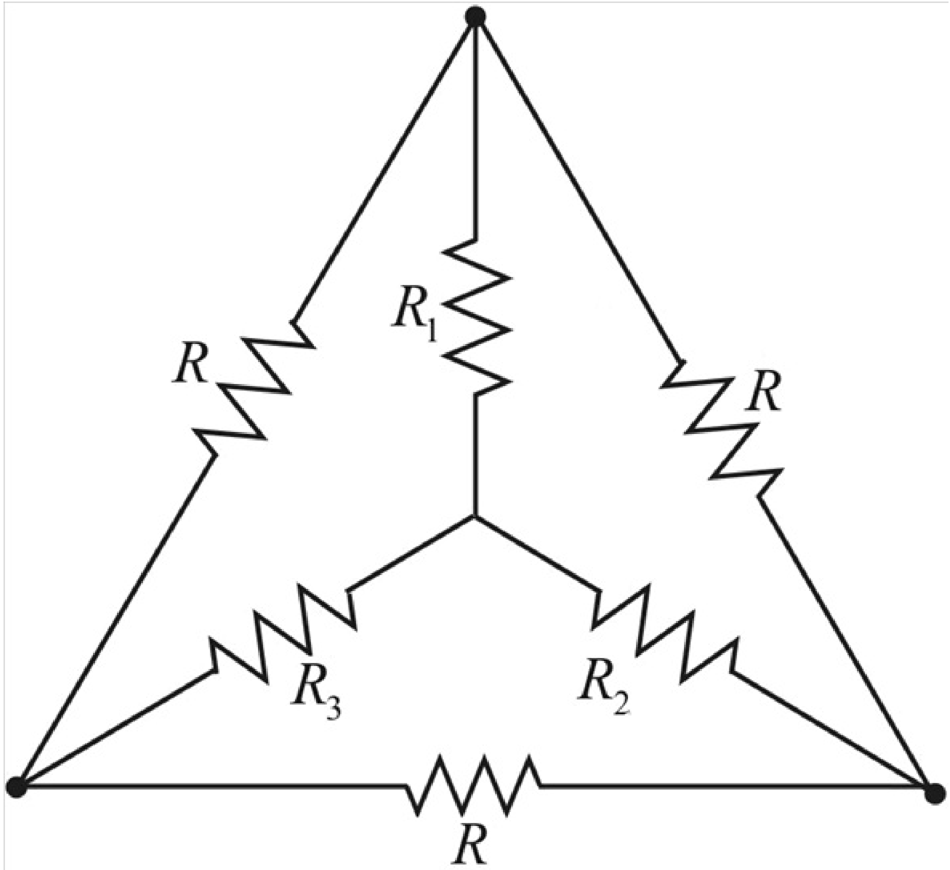
Y - Δ transformation.

In this circuit, there are two Y networks and two Δ networks. Transforming just one of these will simplify the circuit.

Here we convert the upper Δ network into Y network.

Step 3 of 10

The Y equivalent network is,



Step 4 of 10

Each resistor in the Y network is the product of the resistors in the two adjacent Δ branches, divided by the sum of the three Δ resistors.

$$R_1 = \frac{R \times R}{R + R + R}$$

$$R_1 = \frac{R^2}{3R}$$

$$R_1 = \frac{R}{3}$$

Step 5 of 10

$$R_2 = \frac{R \times R}{R + R + R}$$

$$R_2 = \frac{R^2}{3R}$$

$$R_2 = \frac{R}{3}$$

Step 6 of 10

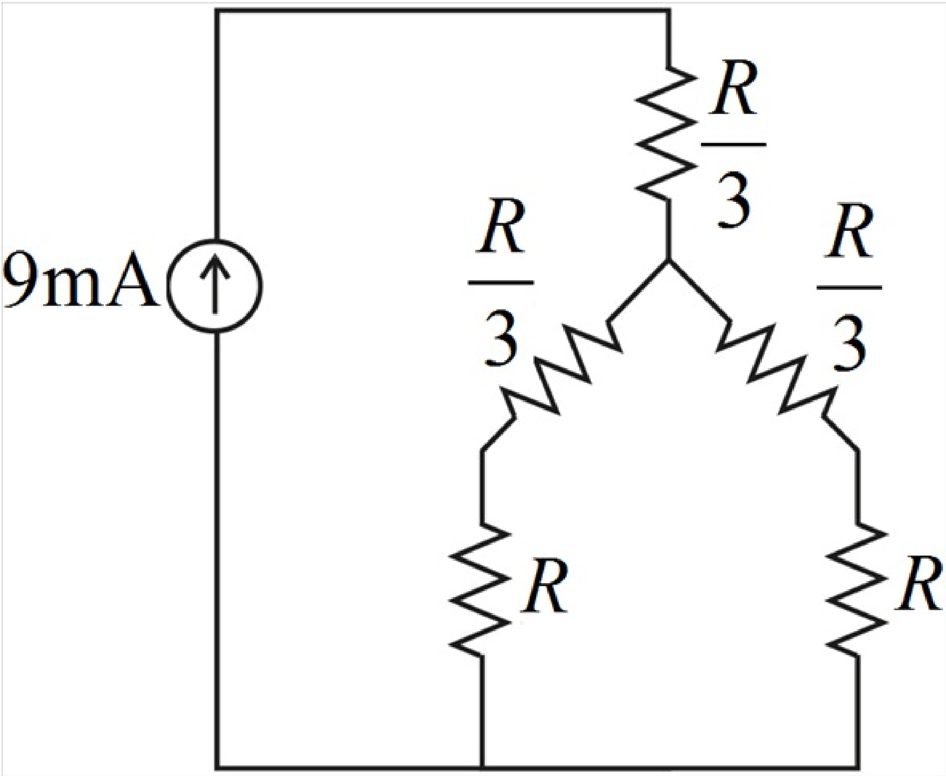
$$R_3 = \frac{R \times R}{R + R + R}$$

$$R_3 = \frac{R^2}{3R}$$

$$R_3 = \frac{R}{3}$$

Step 7 of 10

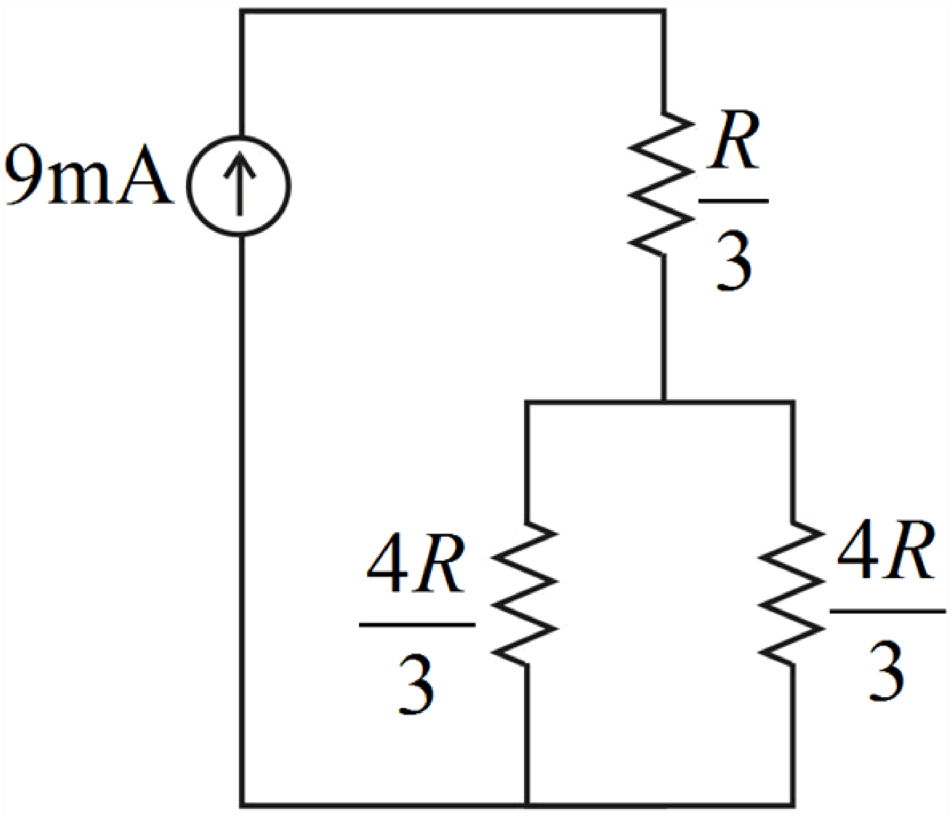
With the Δ converted to Y, the equivalent circuit is as shown below,



Step 8 of 10

Resistors R and $\frac{R}{3}$ are in series, so their combined resistance is

$$R+\frac{R}{3}=\frac{4R}{3}$$



Step 9 of 10

$\frac{4R}{3}$ and $\frac{4R}{3}$ resistors are in parallel, so their combined resistance is

$$\frac{4R}{3}\|\frac{4R}{3}=\frac{\left(\frac{4R}{3}\right)\left(\frac{4R}{3}\right)}{\frac{4R}{3}+\frac{4R}{3}}$$

$$\frac{4R}{3}\|\frac{4R}{3}=\frac{2R}{3}$$

Step 10 of 10

Now this is in series with $\frac{R}{3}$ resistor, so their equivalent resistance is

$$R_{eq}=\frac{2R}{3}+\frac{R}{3}$$

$$R_{eq}=\frac{2R+R}{3}$$

$$R_{eq}=\frac{3R}{3}$$

$$\therefore \boxed{R_{eq}=R}$$